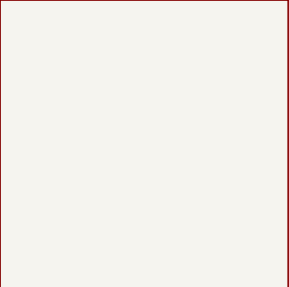
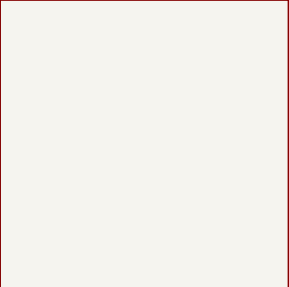


Microcontroller



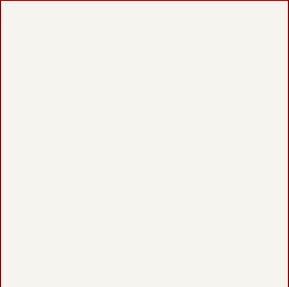
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Sensors



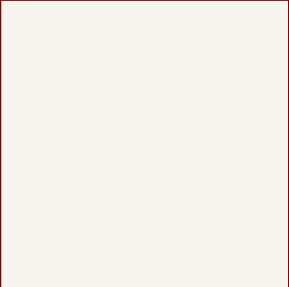
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RF

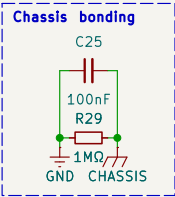


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Power



File: power.kicad_sch

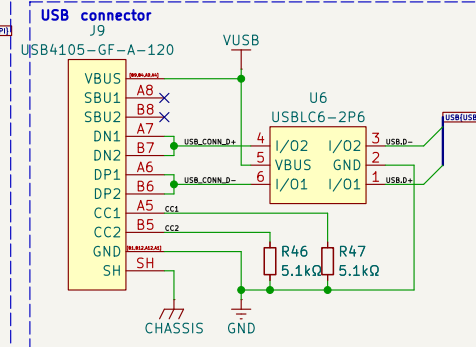
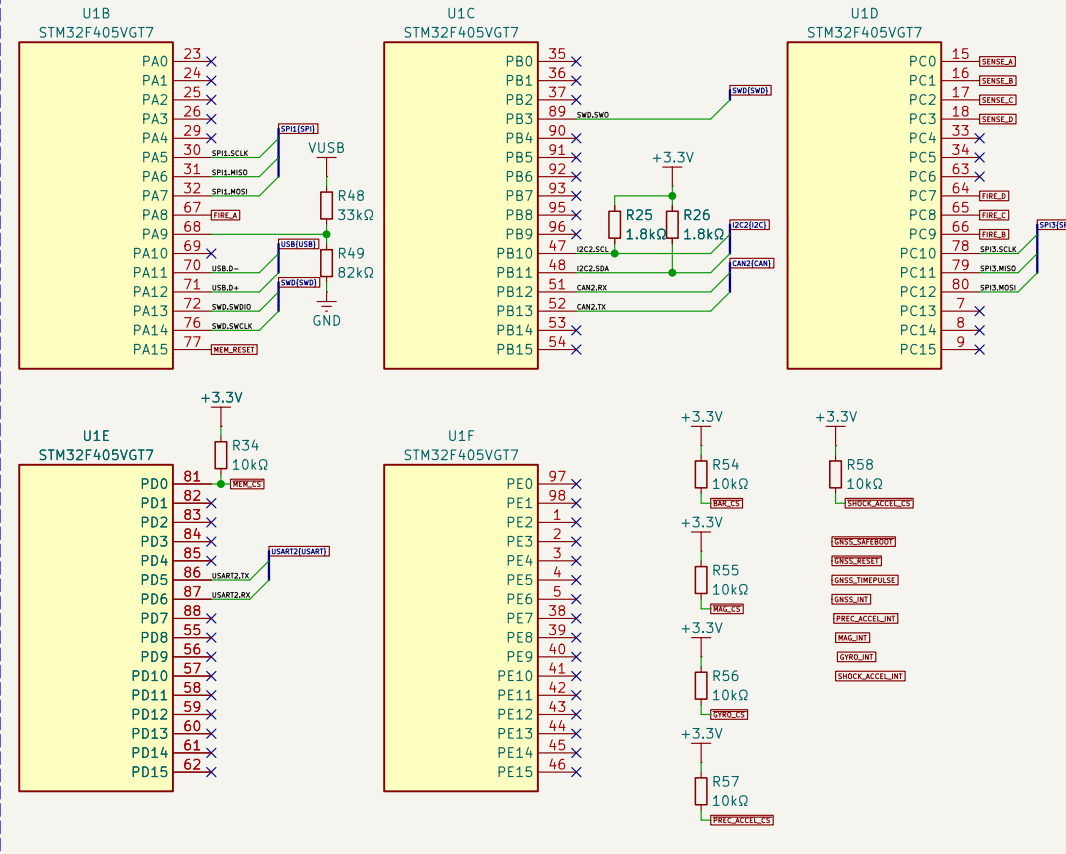


Organization: Lapin Rocketry
Designer: Gianluca Gaiardi
Reviewer: Gianluca Gaiardi
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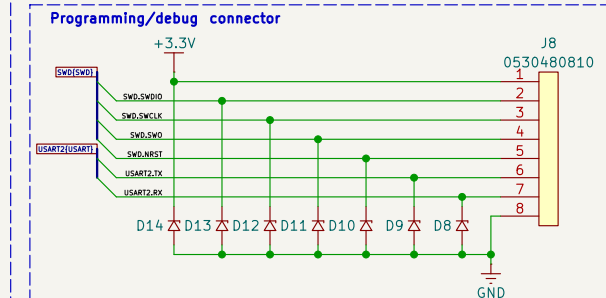


Sheet: / File: flight_computer.kicad_sch		
Flight Computer		
Size: A4	Date: 19/04/2026	Revision: A
KiCad E.D.A. 10.0.3	Sheet 1 of 5	

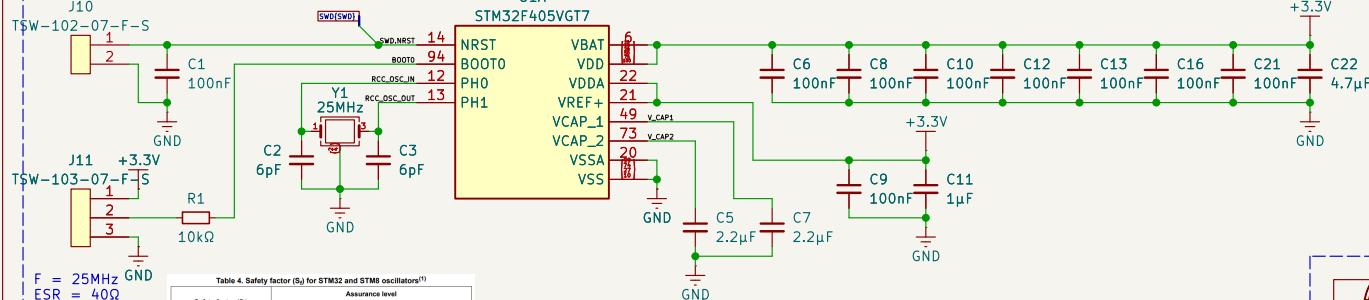
Microcontroller bank A, B, C, D, E



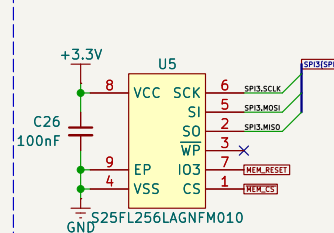
Buzzer
PS1240P02BT



Microcontroller core



Memory



F = 25MHz
ESR = 40Ω
C_L = 8pF
C₀ = 7pF
C_s = 5pF
g_m = 5 mA/V
C_{tot} = C_L + (C_s / 2)

Safety factor (S _f)	Assurance level
S _f ≥ 5	Safe
3 ≤ S _f < 5	Not safe
S _f < 3	Not safe

C_{L2} = 2 × (C_L - C_s) = 6pF
DL = (ESR × (π × F × C_{tot})² × (V_{DD})²) / 2 = 97.07μW
g_m critical = 4 × (ESR) × (2 × π × F)² × (C₀ + C_L)² = 0.89 mA/V
g_m margin = g_m margin / g_m critical = 5.63



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Sheet: /Microcontroller/
File: microcontroller.kicad_sch

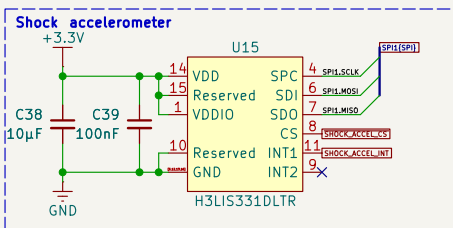
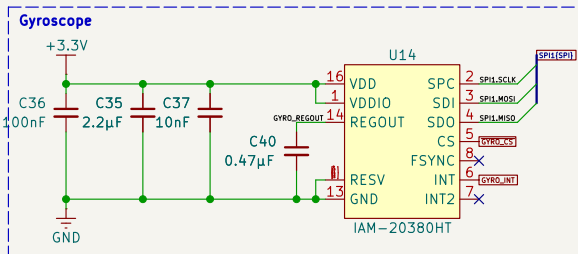
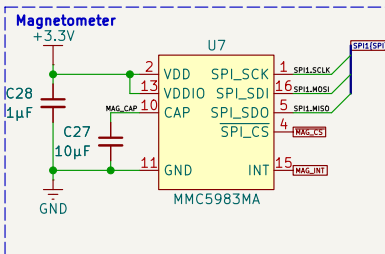
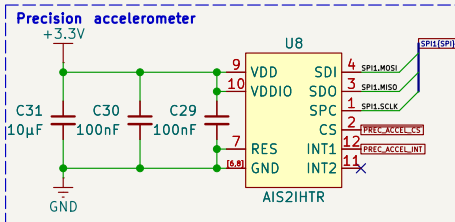
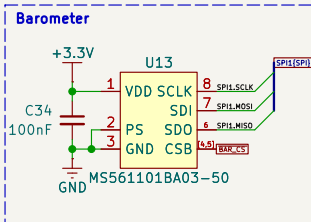
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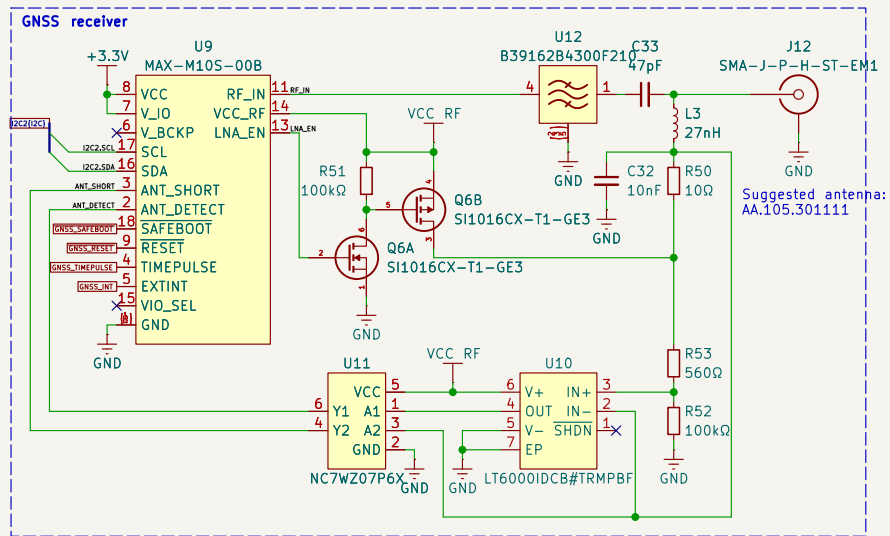
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Revision:

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Sheet 3 of 5





Radio
CC1200
CC1201